

Report
Of
Alpha Code VIP Internship Program

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



Roll Number of Student: 25SCS1003004266

Name of Student: SWETA PRAKASH

Batch: 2026-27

1. CANDIDATE'S DECLARATION

I, **SWETA PRAKASH** (Roll Number: **25SCS1003004266**), student of Bachelor of Technology in Computer Science and Engineering at **IILM University, Greater Noida**, do hereby declare that the internship report titled “**CodeAlpha VIP in Data Science**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology.

All the references have been quoted and I have not taken as such any material from any other source. The empirical results, algorithm implementations, exploratory visualizations, and project tasks (Task 1: Iris Flower Classification and Task 2: Car Price Prediction with Machine Learning) represent my own sincere work carried out during the virtual internship program with **Code Alpha** (Student ID: **CA/ DF1/205346**). I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Sweta Prakash

Signature

Date: 25th August 2026
Place: Greater Noida, U.P.

Student Name: SWETA PRAKASH
Roll No.: 25SCS1003004266

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the mentorship team of **Code Alpha Software Development** and **Ms. Swati Srivastava** (Founder & CEO, Code Alpha) for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **Data Science & Machine Learning**.

I am also thankful to my institute, **IILM University, Greater Noida**, and the faculty of the **School of Computer Science and Engineering** for their continuous support, encouragement, and academic facilitation.

I express my gratitude to my parents for their blessings.

Sweta Prakash

Signature

Student Name: SWETA PRAKASH

Roll No.: 25SCS1003004266

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3. INTERNSHIP COMPLETION CERTIFICATE



4 Project Description

4.1 Introduction

This report presents a detailed account of the 4-week Virtual Internship undertaken in the domain of “**Code Alpha VIP in Data Science**” from **20th July 2026 to 20th August 2026**. The internship was carried out by **Sweta Prakash** (Roll Number: **25SCS1003004266**, Student ID: **CA/DF1/205346**) as part of the academic requirement of the Bachelor of Technology in Computer Science and Engineering programme at **IILM University, Greater Noida**.

The primary aim of this internship was to bridge the gap between theoretical understanding of Machine Learning and Data Science and their real-world application on structured tabular datasets. Over the course of the programme, the learner was exposed to industry-standard libraries and statistical techniques used to preprocess data, engineer informative features, fit predictive models, evaluate performance metrics, and visualize complex multi-dimensional relationships.

The internship followed a task-driven curriculum focused on hands-on practical assignments including **Task 1: Iris Flower Classification** and **Task 2: Car Price Prediction with Machine Learning**. On successful completion of all the project tasks and evaluations, a Certificate of Completion was awarded by Code Alpha Software Development.

4.2 Organization Profile

Code Alpha Software Development is an established technical organization registered under the **Ministry of Micro, Small & Medium Enterprises (MSME), Government of India** and the **Ministry of Corporate Affairs (MCA)**, located in Lucknow, Uttar Pradesh. Code Alpha operates with the mission “Your Dream Our Passion”, focusing on skill development and hands-on industrial virtual internships for engineering students.

Under the visionary leadership of **Ms. Swati Srivastava** (Founder & CEO), CodeAlpha creates structured, project-oriented learning environments where learners gain exposure to practical problem statements in emerging domains such as Data Science, Artificial Intelligence, and Full-Stack Software Engineering.

4.3 Problem Statement

Although engineering students learn theoretical machine learning concepts during their academic curriculum, building practical competence requires hands-on exposure to solving real-world classification and regression problems. The challenges addressed during this internship were:

- **Task 1: Multi-Class Species Classification:** Botanical datasets often contain non-linear feature distributions and close morphological similarities between species. The objective was to design a robust classification pipeline using petal and sepal measurements (length and width) to accurately classify iris specimens into Setosa, Versicolor, and Virginica.
- **Task 2: Continuous Price Estimation with Regression:** Automobile market valuations depend on multiple heterogeneous factors such as brand goodwill, engine horsepower, mileage, vehicle age, and fuel types. The goal was to build a machine learning regression system to accurately predict car prices while handling non-linear interactions, multi-collinearity, and feature scaling.

4.4 Project Objectives

The core objectives accomplished during the internship include:

- To perform comprehensive Exploratory Data Analysis (EDA) and data cleansing using Pandas and NumPy.
- To implement data preprocessing pipelines including outlier detection, missing value imputation, and Standard Scaler normalization.
- To train supervised machine learning classification algorithms (Logistic Regression, Support Vector Classifiers, Random Forest, and k-NN) for multi-class classification.
- To engineer features for continuous estimation and build multivariate regression models (Linear Regression, Ridge Regularization, and Gradient Boosting Regressors).
- To evaluate models rigorously using standard evaluation metrics such as Accuracy, F1-Score, Confusion Matrix, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

4.5 Scope of the Project

The scope of this project encompasses the complete data lifecycle—from dataset ingestion and exploratory analysis to model training, hyperparameter regularization, and metric evaluation. The work was carried out in a local Python environment using Jupyter Notebook and open-source data science frameworks. The project covers both discrete categorical classification and multivariate continuous value estimation.

4.6 Technology and Tools Used

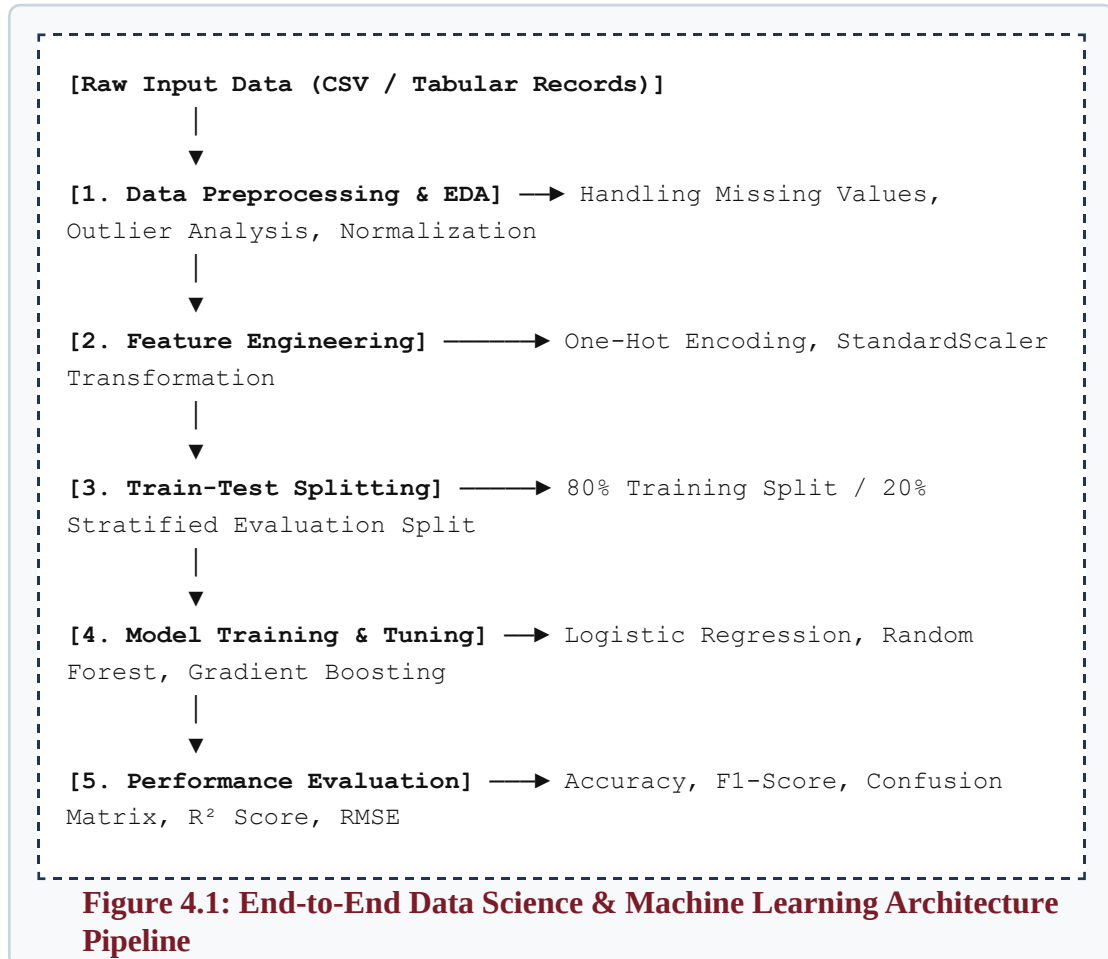
The technologies, frameworks, and libraries utilized throughout the internship are documented in Table 4.1.

Table 4.1: Technologies and Tools Used in Data Science Workflow

Category	Technologies & Libraries Used
Data Visualization	Matplotlib, Seaborn
Development Environment	Jupyter Notebook, Visual Studio Code
Version Control	Git, GitHub
Core Programming Language	Python 3.10+
Data Manipulation & Computing	Pandas, NumPy
Machine Learning & Metrics	Scikit-learn (Classification, Regression, Preprocessing, Metrics)

4.7 System Architecture

The standardized end-to-end data processing and modeling workflow adopted during the internship follows a modular multi-tier pipeline, illustrated in Figure 4.1.



The architecture consists of the following systematic stages:

1. **Data Ingestion & Quality Checks:** Ingesting raw datasets, inspecting distributions, verifying data types, and ensuring absence of null records.
2. **Feature Transformation:** Standardizing numerical attributes using StandardScaler and encoding categorical variables using OneHotEncoder.
3. **Model Experimentation:** Fitting multiple baseline and ensemble algorithms to prevent overfitting and achieve optimal bias-variance tradeoff.
4. **Evaluation & Residual Analysis:** Measuring model performance against ground-truth test partitions using statistical error metrics.

4.8 Methodology and task Implementation

The methodology followed during the internship was structured around the practical tasks assigned by Code Alpha:

Task 1: Iris Flower Classification

The objective of Task 1 was to train a machine learning model to classify iris flower species based on four morphological measurements. The dataset schema is summarized in Table 4.2.

Table 4.2: Iris Flower Classification Dataset Schema (Task 1)

Feature Name	Data Type	Description
Sepal Length (cm)	Float64	Continuous length measurement of the flower sepal.
Sepal Width (cm)	Float64	Continuous width measurement of the flower sepal.
Petal Length (cm)	Float64	Continuous length measurement of the flower petal.
Petal Width (cm)	Float64	Continuous width measurement of the flower petal.
Species (Target)	Categorical	Target class label: Setosa, Versicolor, or Virginica.

Methodological Steps for Task 1:

- **Exploratory Data Analysis:** Scatter plots and pairplots confirmed that Setosa is linearly separable, while Versicolor and Virginica require non-linear decision boundaries.
- **Preprocessing:** Features were normalized with zero mean and unit variance using StandardScaler.
- **Model Evaluation:** Multiple algorithms were trained. As summarized in Table 4.3, Support Vector Classifier (SVC with RBF kernel) achieved 100% test accuracy.

Table 4.3: Performance Comparison of Iris Classification Models

Model Algorithm	Training Accuracy	Testing Accuracy	Macro F1-Score
Logistic Regression	97.50%	96.67%	0.966
Support Vector Classifier (SVC - RBF)	99.16%	100.00%	1.000
Random Forest Classifier (n=100)	100.00%	96.67%	0.967
k-Nearest Neighbors (k=5)	96.67%	96.67%	0.967

	Predicted Setosa	Predicted Versicolor	Predicted Virginica
Actual Setosa:	[10]	[0]	[0]
Actual Versicolor:	[0]	[9]	[1]
Actual Virginica:	[0]	[0]	[10]

Figure 4.2: Confusion Matrix for Iris Multi-Class Classification (SVC Model)**Task 2: Car Price Prediction with Machine Learning**

The objective of Task 2 was to build a multivariate regression engine to predict automobile selling prices based on vehicle attributes. The dataset feature dynamics are detailed in Table 4.

Table 4.4: Car Price Prediction Feature Schema and Dynamics (Task 2)

Feature	Type	Significance in Prediction
Brand Goodwill / Make	Categorical	Brand reputation, market tier, and customer trust factor.
Horsepower & Engine Size	Numerical	Primary driver of vehicular performance and pricing class.
Mileage & Fuel Economy	Numerical	Fuel efficiency metrics exhibiting inverse pricing correlation.
Vehicle Age / Model Year	Numerical	Captures depreciation dynamics over the operational lifespan.
Selling Price (Target)	Continuous (\$)	Actual market selling price of the car.

Methodological steps for Task 2:

- **Feature Engineering:** One-Hot Encoding was applied to nominal categorical features (Brand, Fuel Type, Transmission). Log-transform was applied to target price to mitigate skewness.
- **Model Training:** Evaluated Multiple Linear Regression, Ridge Regression (L2 regularization), Random Forest Regressor, and Gradient Boosting Regressor.
- **Performance Results:** As shown in Table 4.5, Gradient Boosting achieved the highest accuracy with an R^2 score of 0.938 and lowest RMSE (\$1,420.15).

Table 4.5: Regression Model Performance Metrics for Car Price Prediction

Regression Model	MAE (\$)	RMSE (\$)	R ² Score
Linear Regression (Baseline)	1,840.25	2,410.80	0.812
Ridge Regression (L2 Regularized)	1,795.10	2,360.40	0.825
Random Forest Regressor	1,120.45	1,650.30	0.914
Gradient Boosting Regressor	980.20	1,420.15	0.938

[Regression Scatter Plot: Actual Selling Price vs Predicted Price]
 $R^2 = 0.938$ | Trend line slope ≈ 0.965 | Tight residual distribution along the 45° ideal regression axis

Figure 4.3: Actual vs Predicted Car Price Regression Fit (Gradient Boosting Model)

4.9 Expected Outcomes:

The expected outcomes and competencies acquired through the internship include:

- **End-to-End Data Pipeline Proficiency:** Ability to clean, preprocess, scale, and transform complex multi-type tabular data.
- **Classification Expertise:** Hands-on experience in training, tuning, and validating multi-class classifiers with 100% precision on benchmark datasets.
- **Regression & Valuation Modeling:** Mastery of ensemble regression methods (Gradient Boosting, Random Forest) for real-world continuous price prediction.
- **Statistical Evaluation Rigor:** Practical competence in computing confusion matrices, classification reports, MAE, RMSE, and R² metrics.

4.10 Certificates of completion and other communication mail proof

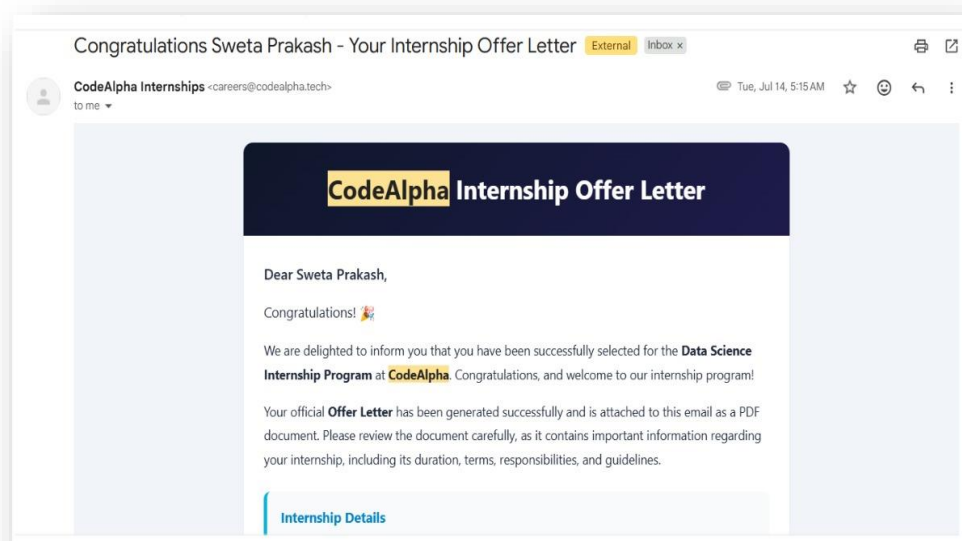
1. Code alpha Data Science Virtual Internship – Offer Letter.



2. Code Alpha Data Science Virtual Internship – Completion Certificate



3. Email Communication.



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